Regional brain volumes from T1 MRI - Insight46 phase 1

Patrick Esser

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Document details

Categories of variables	Instrumented 10metre Gait assessment
Summary of work undertaken	Cleaning / Derivation / **Both±** Description/Rationale: Derivation: Conversion from Raw 3D accelerometer & quaternions data to temporal, spatial and non-linear outcome measurements Cleaning: Data check for missing data, insufficiently available data (e.g. too little steps, duration too short, not within 'norm' parameters, etc) and adjustment for 'double peak syndrome' prevalent in non-heelstrike walkers.
Output variables	I46_Gait

Documentation – I46_GAIT – Insight phase 1

| **List any papers in which cleaned/derived variables have been used** | **Related to Insight46:** Tektonidis, TG., Coe, S., Esser, P., Maddock, J., Buchanan, S., Mavrommati, F., Schott, JM., Izadi, H., Richards, M., Dawes, H., Longitudinal Changes in Diet Quality over Adulthood and Physical Function in Older Life: Findings from a British Birth Cohort, Current Developments in Nutrition, 2020 (Suppl 2) Tektonidis, TG., Coe, S., Esser, P., Maddock, J., Buchanan, S., Mavrommati, F., Schott, JM., Izadi, H., Richards, M., Dawes, H., Diet quality in late midlife is associated with faster walking speed in later life in women, but not men: findings from a prospective British birth cohort British Journal of Nutrition, 2019, (1-21) Tektonidis TG, Esser P, Coe S, Maddock J, Buchanan S, Mavrommati F, Schott JM, Izadi H, Richards M, Dawes H., MON-LB692: Diet Quality in Late Midlife is Associated with Faster Walking Speed in Later Life in Women, but Not Men: Findings From a British Birth Cohort; Clinical Nutrition 38 (S316-S317), 2019 |

Information

Excessive number of NaNs in two variables:

The variables & have a greater number of NaN available. This is due to the delicate analysis whereby there needs to be a clear heelstrike, midfoot and toe-off for both sides for a minimum of 5 steps present, before the values can be derived. In the elderly cohort, this may not always be the case over each walk, hence the number of NaNs in these and associated standard deviation values may be significantly more prevalent.

Note re. difference in analysis from phase 1 (added by Sarah Keuss 26/6/24):

Sarah Keuss noted that there were results for two walks per task in phase 1 but not in phase 2. Heidi Murray-Smith contacted Patrick Esser (who analysed the phase 2 data) and Thanasis Tektonidis (Patrick Esser's PhD student who worked on the phase 1 data) and the following explanation was given. In phase 1, the data was analysed as two separate walks: walk 1, the 10-metre walk towards the cone; and walk 2, the 10-metre walk back from the cone. In phase 2, a newer version of the analysis was used which has the automaticity to exclude the turn at the cone. As such, there is a combined walk per task which is an average of the walk to the cone and the walk back from the cone.

Key publications

Not related to Insight 46 (including Development, Deployment and clinical utilisation):

Cinar, E., Weedon, B., Esser, P., Joshi, S., Delextrat, A.,Collett, J., Springett, D.N., Dawes, H., Dual-task effect on gait in healthy adolescents: Association between health-related indicators and DT performance, Journal of Motor Behaviour, 2020

Muhaidat J., Al-Yahya E., Mohammad M.T., Qutishat D., Okasheh R., Al-Khlaifat L., Lawrie S., Esser P., Dawes H., Dual-tasking in older women: physical activity or else?, Journal of Women & Aging, 2020 (1-11)

Xu, L., Zeng, H., Shen, W., Zhao, J., Shao, M., Zhao, J, Chen, H., Hu, H., Esser, P., Dawes H., Liu, F., The impact of lower extremity neuropathy and artery disease on gait characteristics in type 2 diabetic individuals, Research Square, 2020, DOI: 10.21203/rs.3.rs-28545

Mansoubi, M., Weedon, B.D., Esser, P., Mayo, N., Fazel, M., Wade, W., Ward, E.W., Kemp, S., Delextrat, A., Dawes, H., Cognitive performance, quality and quantity of movement reflect psychological symptoms in adolescents, Journal of Sports Science & Medicine, 2020

Al-Yahya, E., Mohammad M.T., Muhaidat J., Al Demour, S., Qutishat, D., Al-Khalaifat, L., Okasheh, R., Esser, P., Dawes, H., Functional balance and gait characteristics in men with lower urinary tract symptoms secondary to benign prostatic hyperplasia. American Journal of Mens Health, Feb 2019

Esser P., Maynard K., Collet J., Steins D., Hillier A., Buckingham J., Tan G., King L., Dawes H., Single sensor gait analysis as an indicator for diabetic peripheral neuropathy: A Proof of principle study. Diabetes & Metabolism Journal, 2017

Cuzzolin, F., Sapienza, M., Esser, P., Saha, S., Franssen, M., Collett, J., Dawes, H. Metric learning for Parkinsonian identification from IMU gait measurements¸ Gait & Posture, 2017, 54 · February 2017

Collett J, Esser P, Khalil H, Busse M, Quinn L, DeBono K, et al. Insights into gait disorders: Walking variability using phase plot analysis, Huntington's Disease. Gait & Posture. 2014. 40(4): p694-700

Steins D, Sheret I, Dawes H, Esser P, Collett J. A smart device inertial-sensing method for gait analysis. Journal of Biomechanics. 2014. 47(15): p.3780-5

Esser, P, Dawes, H., Collett J., Howells, K., Insights into gait disorders: walking variability using phase plot analysis, Parkinson's disease. Gait Posture, 2013. 38(4): p. 648-52.

Esser P, Dawes H, Collett J, Feltham MG, Howells K. Validity and inter-rater reliability of Inertial Gait Measurements in Parkinson's Disease: A pilot study J Neurolog Sci 2012. 205(1): p. 177-81

Esser P, Dawes H, Collett J, Feltham MG, Howells K. Assessment of spatio-temporal gait parameters using inertial measurement units in neurological populations Gait & Posture 2011. 34(4): p. 558-60

Esser P, Dawes H, Collett J, Howells K. IMU: Inertial Sensing of Vertical CoM Movement. Journal of Biomechanics 2009;42:1578 - 81.

Variables in this document

336 high-confidence matches

This list is very large (336 variables) and is not reproduced in this PDF. Please visit the OWL website to see the full list.

2 low-confidence matches

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]		
back	In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Physical capability [5108]		
walk	Do you find it difficult to walk for a quarter of a mile on the level because of long term health problems? - at age 53 years	prose scan